WEEK 3 PROJECT REPORT

PROCESS SCHEDULING SIMULATOR

Design and Implementation of CPU Scheduling Algorithms Using C

Project-Based Learning (PBL)

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Technologies Used:
C Programming, GCC, Linux, Make, Git and GitHub

Project Repository:
PBL-kiri

Academic Year:
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1. Introduction

Process scheduling is an important function of an operating system. It determines the order in which processes are executed by the CPU. Different scheduling algorithms use different methods to allocate CPU time to processes.

The Process Scheduling Simulator is developed using the C programming language to demonstrate and compare different CPU scheduling algorithms. It accepts process details such as Process ID, Arrival Time, Burst Time, and Priority, and displays the execution order using a Gantt chart.

The simulator calculates performance metrics such as Waiting Time, Turnaround Time, Response Time, and CPU Utilization to help analyze the performance of each algorithm.

2. Objectives

- To implement different CPU scheduling algorithms using C.
- To understand the working of preemptive and non-preemptive scheduling.
- To display process execution using Gantt charts.
- To calculate Waiting Time, Turnaround Time, and Response Time.
- To calculate CPU Utilization.
- To compare the performance of scheduling algorithms using different test cases.

3. Input Specification

THE SIMULATOR ACCEPTS THE FOLLOWING DETAILS FOR EACH PROCESS:

Input	Description
Process ID (PID)	Unique identifier assigned to each process
Arrival Time (AT)	Time at which the process enters the ready queue
Burst Time (BT)	Total CPU time required by the process
Priority (PR)	Priority assigned to each process
Time Quantum	Maximum CPU time allocated to a process in one turn of Round Robin scheduling

The user enters the process details, selects a scheduling algorithm, and views the Gantt chart and calculated performance metrics.

4. Performance Metrics and Formulas

1. Completion Time (CT)

The time at which a process finishes execution.

2. Turnaround Time (TAT)

TAT = Completion Time − Arrival Time

3. Waiting Time (WT)

WT = Turnaround Time − Burst Time

4. Response Time (RT)

RT = First CPU Start Time − Arrival Time

5. Average Waiting Time

Average WT = Sum of Waiting Times ÷ Number of Processes

6. Average Turnaround Time

Average TAT = Sum of Turnaround Times ÷ Number of Processes

7. CPU Utilization

CPU Utilization = (CPU Busy Time ÷ Total Elapsed Time) × 100%

5. Scheduling Algorithms

The Process Scheduling Simulator implements six CPU scheduling algorithms. Each algorithm follows a different method to select processes for CPU execution.

5.1 First Come First Serve (FCFS)

FCFS is a non-preemptive scheduling algorithm in which processes are executed in the order they arrive in the ready queue. Once a process starts execution, it continues until completion.

5.2 Shortest Job First (SJF) — Non-Preemptive

SJF Non-Preemptive selects the available process with the shortest burst time. Once selected, the process executes until it finishes.

5.3 Shortest Remaining Time First (SRTF) — Preemptive

SRTF is the preemptive version of SJF. The CPU executes the process with the shortest remaining burst time. If a process with a shorter remaining time arrives, the currently running process may be interrupted.

5.4 Priority Scheduling — Non-Preemptive

Priority Non-Preemptive selects the available process with the highest priority. Once execution begins, the process continues until completion.

5.5 Priority Scheduling — Preemptive

Priority Preemptive selects the process with the highest priority. If a higher-priority process arrives, it can interrupt the currently running process.

5.6 Round Robin (RR)

Round Robin assigns each process a fixed time slice called the time quantum. If a process does not finish within its allocated time, it returns to the ready queue to wait for another turn.

6. Test Case 1 — Process Scheduling

6.1 Input Details

The first test case is used to evaluate the behavior of the CPU scheduling algorithms using four processes with different arrival times, burst times, and priorities.

Process ID	Arrival Time (AT)	Burst Time (BT)	Priority (PR)
P1	0	7	2
P2	2	4	1
P3	4	1	3
P4	5	2	2

The same input values are used for all six scheduling algorithms so that their performance can be compared fairly.

Test Case 1

6.1 First Come First Serve (FCFS) Scheduling

```
/scheduler
==================================
   PROCESS SCHEDULING SIMULATOR
==================================
Enter number of processes (1-100): 4

Process 1
PID: 1
Arrival Time: 0
Burst Time: 7
Priority (smaller number = higher priority): 2

Process 2
PID: 2
Arrival Time: 2
Burst Time: 4
Priority (smaller number = higher priority): 1

Process 3
PID: 3
Arrival Time: 4
Burst Time: 1
Priority (smaller number = higher priority): 3

Process 4
PID: 4
Arrival Time: 5
Burst Time: 2
Priority (smaller number = higher priority): 2
```

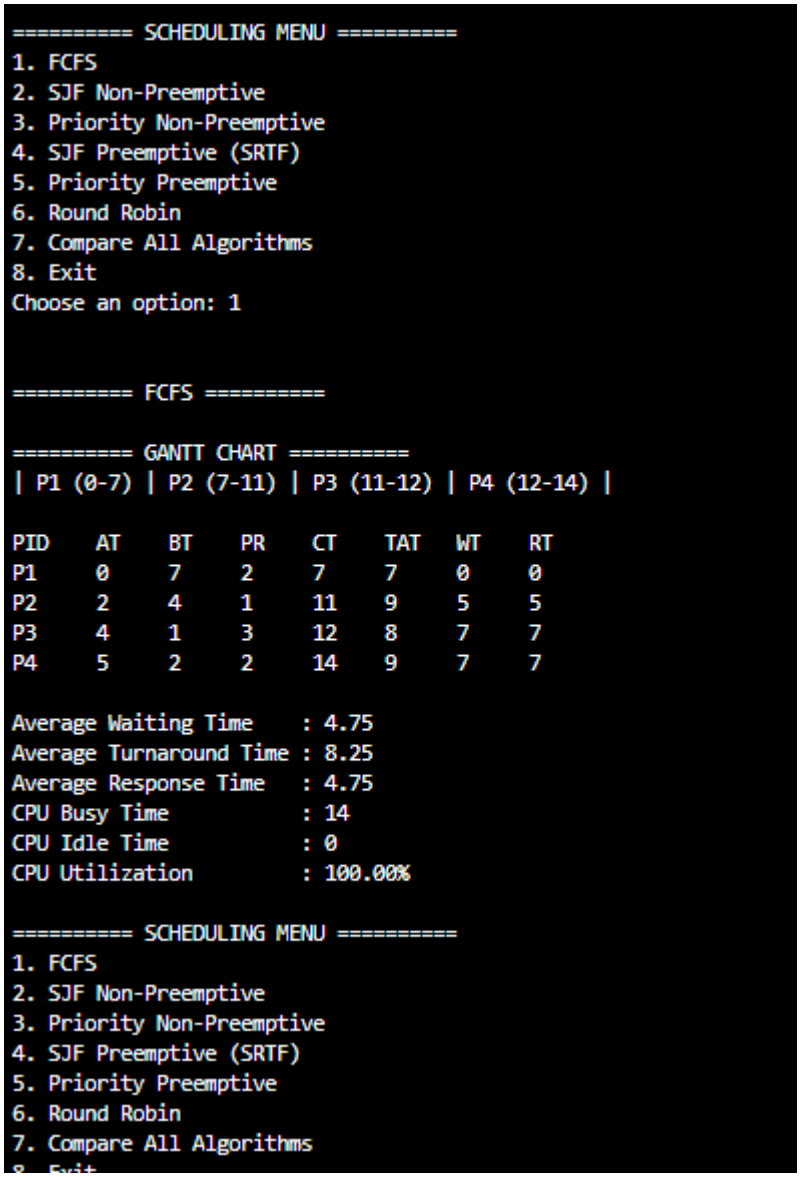
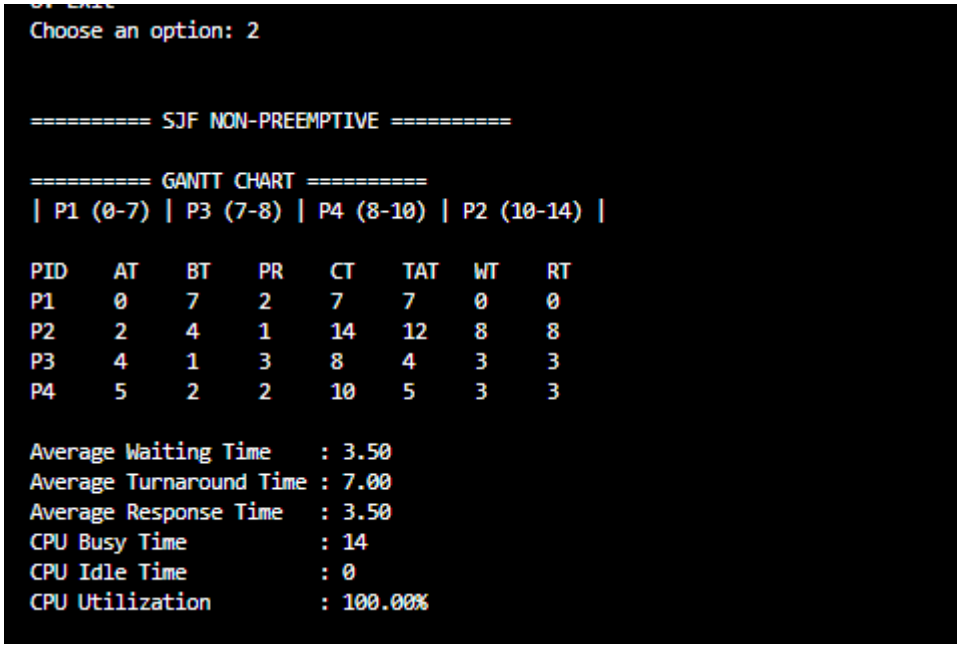


Figure 1: FCFS Scheduling Output for Test Case 1

5.2 Shortest Job First (SJF) — Non-Preemptive



```
Choose an option: 2


========== SJF NON-PREEMPTIVE ==========

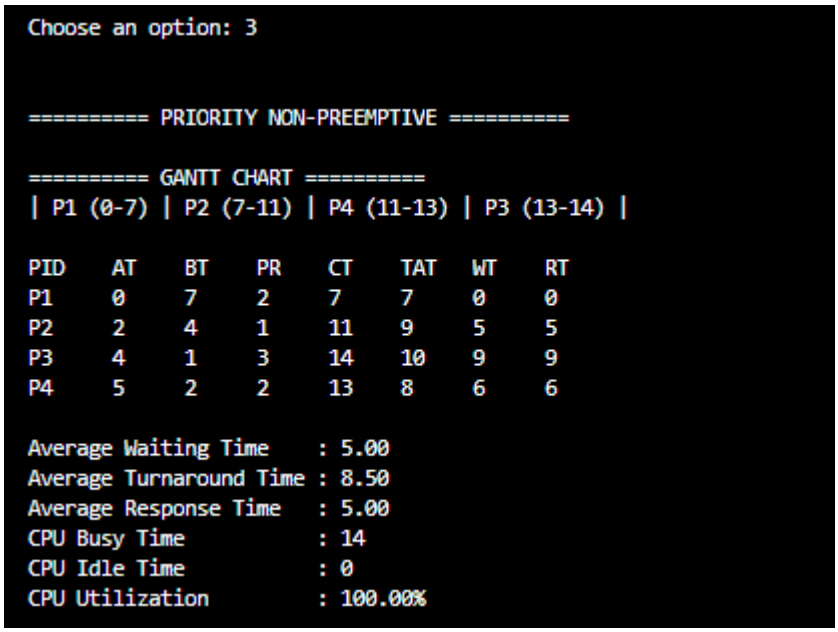
========== GANTT CHART ==========
| P1 (0-7) | P3 (7-8) | P4 (8-10) | P2 (10-14) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     7     2     7     7     0     0
P2     2     4     1     14    12    8     8
P3     4     1     3     8     4     3     3
P4     5     2     2     10    5     3     3

Average Waiting Time    : 3.50
Average Turnaround Time : 7.00
Average Response Time   : 3.50
CPU Busy Time           : 14
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

Figure 2: SJF Non-Preemptive Scheduling Output for Test Case 1.

5.3 Priority Non-Preemptive Scheduling



```
Choose an option: 3


========== PRIORITY NON-PREEMPTIVE ==========

========== GANTT CHART ==========
| P1 (0-7) | P2 (7-11) | P4 (11-13) | P3 (13-14) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     7     2     7     7     0     0
P2     2     4     1     11    9     5     5
P3     4     1     3     14    10    9     9
P4     5     2     2     13    8     6     6

Average Waiting Time    : 5.00
Average Turnaround Time : 8.50
Average Response Time   : 5.00
CPU Busy Time           : 14
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

Figure 3: Priority Non-Preemptive Scheduling Output for Test Case 1

5.4 SJF Preemptive (SRTF)

```
Choose an option: 4


========== SJF PREEMPTIVE (SRTF) ==========

========== GANTT CHART ==========
| P1 (0-2) | P2 (2-4) | P3 (4-5) | P2 (5-7) | P4 (7-9) | P1 (9-14) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     7     2     14    14    7     0
P2     2     4     1     7     5     1     0
P3     4     1     3     5     1     0     0
P4     5     2     2     9     4     2     2

Average Waiting Time    : 2.50
Average Turnaround Time : 6.00
Average Response Time   : 0.50
CPU Busy Time           : 14
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

Figure 4: SJF Preemptive (SRTF) Scheduling Output for Test Case 1.

5.5 Priority Preemptive Scheduling

```
Choose an option: 5


========== PRIORITY PREEMPTIVE ==========

========== GANTT CHART ==========
| P1 (0-2) | P2 (2-6) | P1 (6-11) | P4 (11-13) | P3 (13-14) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     7     2     11    11    4     0
P2     2     4     1     6     4     0     0
P3     4     1     3     14    10    9     9
P4     5     2     2     13    8     6     6

Average Waiting Time    : 4.75
Average Turnaround Time : 8.25
Average Response Time   : 3.75
CPU Busy Time           : 14
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

Figure 5: Priority Preemptive Scheduling Output for Test Case 1.

5.6 Round Robin Scheduling

```
Choose an option: 6
Enter Time Quantum: 2


========== ROUND ROBIN ==========

========== GANTT CHART ==========
| P1 (0-2) | P2 (2-4) | P1 (4-6) | P3 (6-7) | P2 (7-9) | P4 (9-11) | P1 (11-14) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     7     2     14    14    7     0
P2     2     4     1     9     7     3     0
P3     4     1     3     7     3     2     2
P4     5     2     2     11    6     4     4

Average Waiting Time    : 4.00
Average Turnaround Time : 7.50
Average Response Time   : 1.50
CPU Busy Time           : 14
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

Figure 6: Round Robin Scheduling Output for Test Case 1.

Compare All Algorithms

```
Choose an option: 7


========== FCFS ==========

========== GANTT CHART ==========
| P1 (0-7) | P2 (7-11) | P3 (11-12) | P4 (12-14) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     7     2     7     7     0     0
P2     2     4     1     11    9     5     5
P3     4     1     3     12    8     7     7
P4     5     2     2     14    9     7     7

Average Waiting Time    : 4.75
Average Turnaround Time : 8.25
Average Response Time   : 4.75
CPU Busy Time           : 14
CPU Idle Time           : 0
CPU Utilization         : 100.00%


========== SJF NON-PREEMPTIVE ==========

========== GANTT CHART ==========
| P1 (0-7) | P3 (7-8) | P4 (8-10) | P2 (10-14) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     7     2     7     7     0     0
P2     2     4     1     14    12    8     8
P3     4     1     3     8     4     3     3
P4     5     2     2     10    5     3     3

Average Waiting Time    : 3.50
Average Turnaround Time : 7.00
Average Response Time   : 3.50
CPU Busy Time           : 14
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

```
========== PRIORITY NON-PREEMPTIVE ==========

========== GANTT CHART ==========
| P1 (0-7) | P2 (7-11) | P4 (11-13) | P3 (13-14) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     7     2     7     7     0     0
P2     2     4     1     11    9     5     5
P3     4     1     3     14    10    9     9
P4     5     2     2     13    8     6     6

Average Waiting Time    : 5.00
Average Turnaround Time : 8.50
Average Response Time   : 5.00
CPU Busy Time           : 14
CPU Idle Time           : 0
CPU Utilization         : 100.00%


========== SJF PREEMPTIVE (SRTF) ==========

========== GANTT CHART ==========
| P1 (0-2) | P2 (2-4) | P3 (4-5) | P2 (5-7) | P4 (7-9) | P1 (9-14) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     7     2     14    14    7     0
P2     2     4     1     7     5     1     0
P3     4     1     3     5     1     0     0
P4     5     2     2     9     4     2     2

Average Waiting Time    : 2.50
Average Turnaround Time : 6.00
Average Response Time   : 0.50
CPU Busy Time           : 14
CPU Idle Time           : 0
CPU Utilization         : 100.00%


========== PRIORITY PREEMPTIVE ==========

========== GANTT CHART ==========
| P1 (0-2) | P2 (2-6) | P1 (6-11) | P4 (11-13) | P3 (13-14) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     7     2     11    11    4     0
P2     2     4     1     6     4     0     0
P3     4     1     3     14    10    9     9
P4     5     2     2     13    8     6     6

Average Waiting Time    : 4.75
Average Turnaround Time : 8.25
Average Response Time   : 3.75
CPU Busy Time           : 14
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

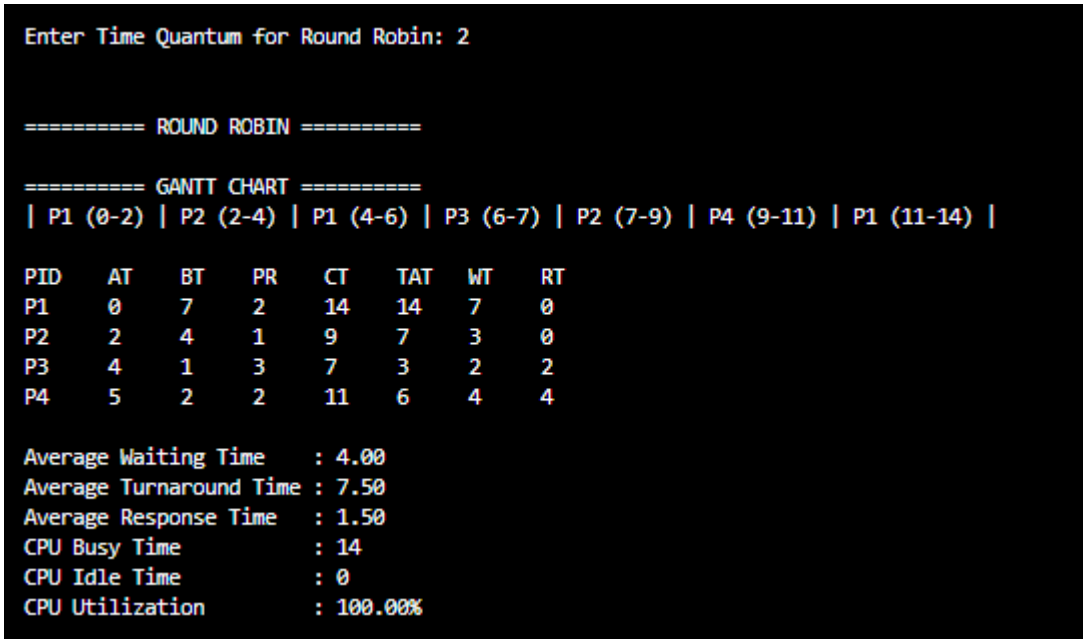


Figure 7: Comparison of CPU Scheduling Algorithms for Test Case 1.

TEST CASE 2

Test Case 2 — Process Inputs

PID	Arrival Time	Burst Time	Priority
P1	0	3	3
P2	1	6	1
P3	2	2	4
P4	4	4	2

5.7 Test Case 2 — FCFS Scheduling

```
8. Exit
Choose an option: 1


========== FCFS ==========

========== GANTT CHART ==========
| P1 (0-3) | P2 (3-9) | P3 (9-11) | P4 (11-15) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     3     3     3     3     0     0
P2     1     6     1     9     8     2     2
P3     2     2     4     11    9     7     7
P4     4     4     2     15    11    7     7

Average Waiting Time    : 4.00
Average Turnaround Time : 7.75
Average Response Time   : 4.00
CPU Busy Time           : 15
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

Figure 8: FCFS Scheduling Output for Test Case 2.

Test Case 2 — SJF Non-Preemptive Scheduling

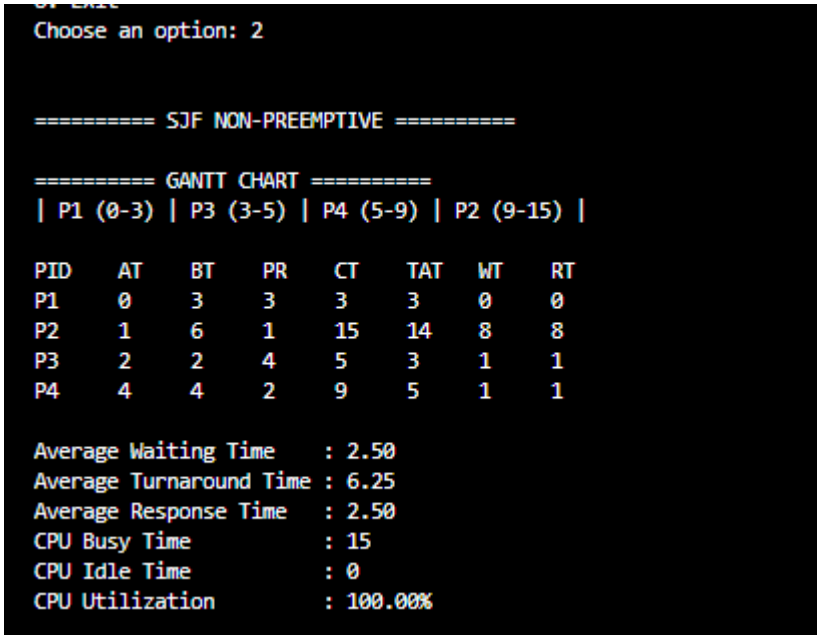


Figure 9: SJF Non-Preemptive Scheduling Output for Test Case 2.

Test Case 2 — Priority Non-Preemptive Scheduling

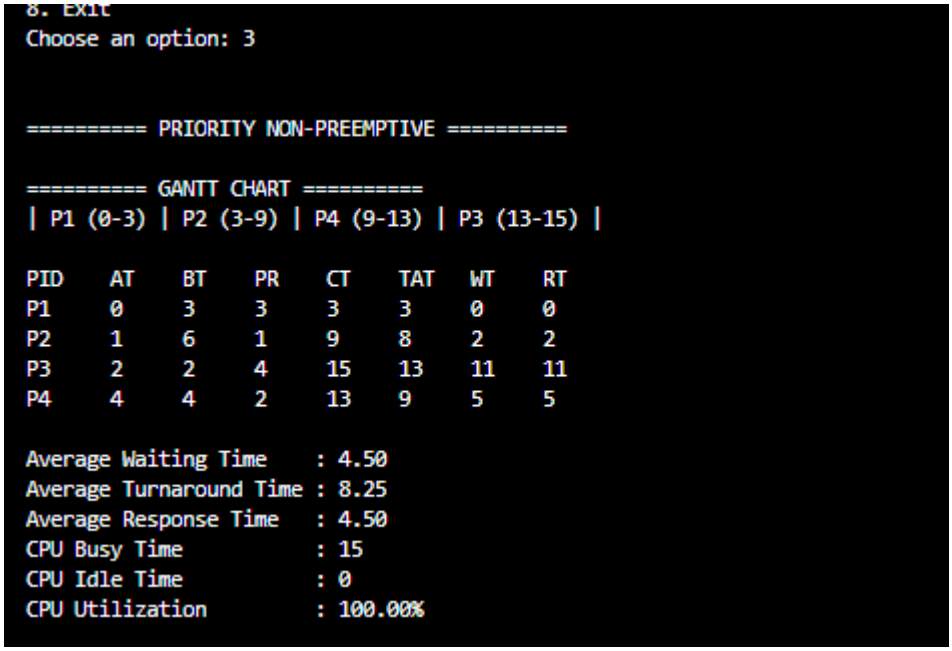


Figure 10: Priority Non-Preemptive Scheduling Output for Test Case 2.

Test Case 2 — SJF Preemptive (SRTF) Scheduling

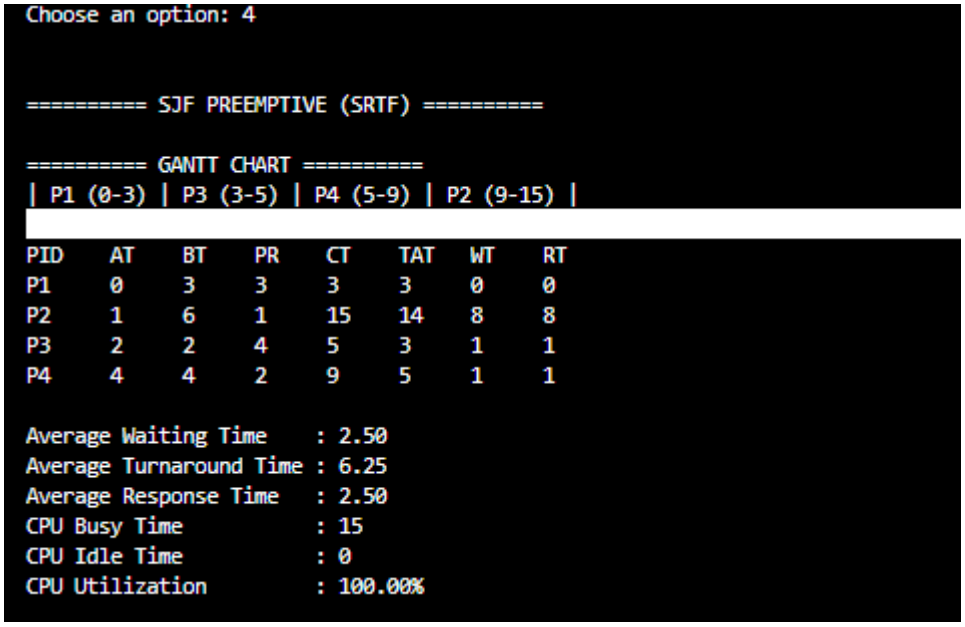


Figure 11: SJF Preemptive (SRTF) Scheduling Output for Test Case 2.

Test Case 2 — Priority Preemptive Scheduling

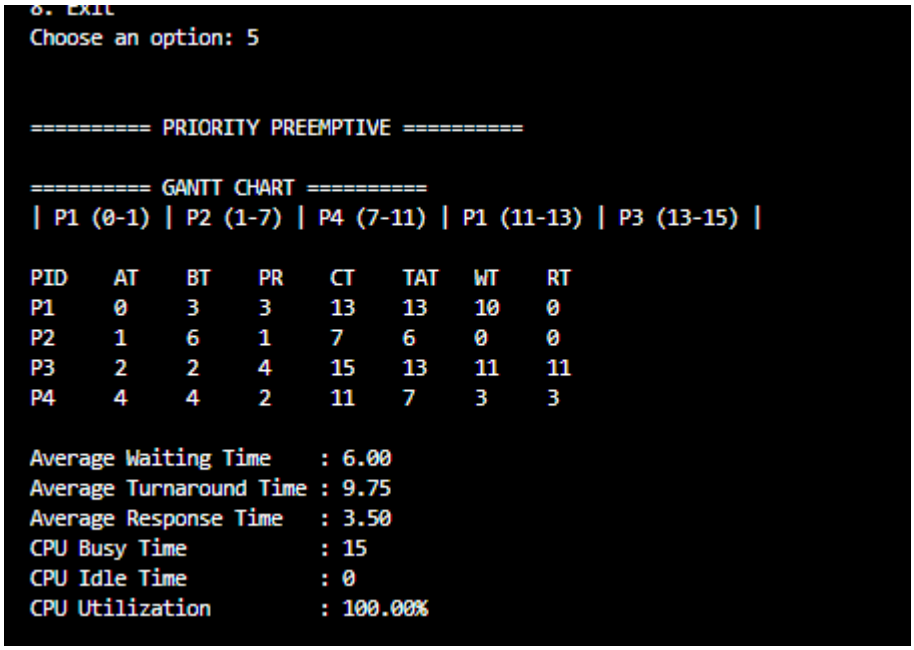


Figure 12: Priority Preemptive Scheduling Output for Test Case 2.

Test Case 2 — Round Robin Scheduling

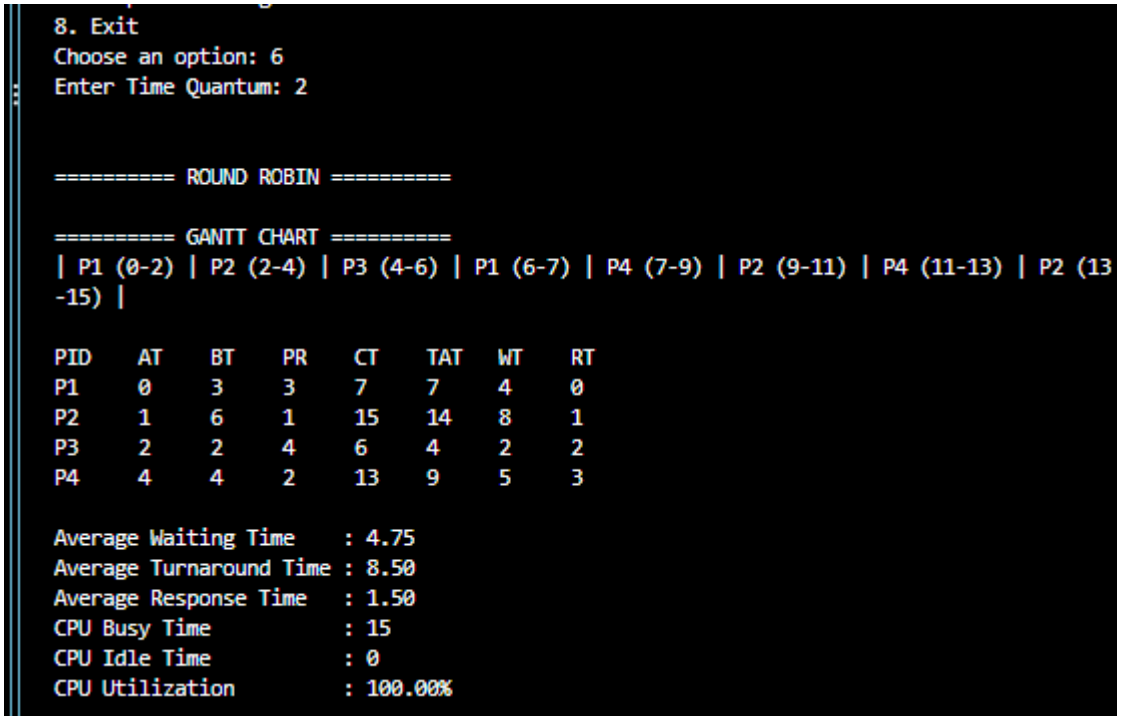


Figure 13: Round Robin Scheduling Output for Test Case 2.

Test Case 2 — Overall Algorithm ComparisoN

```
Choose an option: 7


========== FCFS ==========

========== GANTT CHART ==========
| P1 (0-3) | P2 (3-9) | P3 (9-11) | P4 (11-15) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     3     3     3     3     0     0
P2     1     6     1     9     8     2     2
P3     2     2     4     11    9     7     7
P4     4     4     2     15    11    7     7

Average Waiting Time    : 4.00
Average Turnaround Time : 7.75
Average Response Time   : 4.00
CPU Busy Time           : 15
CPU Idle Time           : 0
CPU Utilization         : 100.00%


========== SJF NON-PREEMPTIVE ==========

========== GANTT CHART ==========
| P1 (0-3) | P3 (3-5) | P4 (5-9) | P2 (9-15) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     3     3     3     3     0     0
P2     1     6     1     15    14    8     8
P3     2     2     4     5     3     1     1
P4     4     4     2     9     5     1     1

Average Waiting Time    : 2.50
Average Turnaround Time : 6.25
Average Response Time   : 2.50
CPU Busy Time           : 15
CPU Idle Time           : 0
CPU Utilization         : 100.00%


========== PRIORITY NON-PREEMPTIVE ==========

========== GANTT CHART ==========
| P1 (0-3) | P2 (3-9) | P4 (9-13) | P3 (13-15) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     3     3     3     3     0     0
P2     1     6     1     9     8     2     2
P3     2     2     4     15    13    11    11
P4     4     4     2     13    9     5     5

Average Waiting Time    : 4.50
Average Turnaround Time : 8.25
Average Response Time   : 4.50
CPU Busy Time           : 15
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

```
========== SJF PREEMPTIVE (SRTF) ==========

========== GANTT CHART ==========
| P1 (0-3) | P3 (3-5) | P4 (5-9) | P2 (9-15) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     3     3     3     3     0     0
P2     1     6     1     15    14    8     8
P3     2     2     4     5     3     1     1
P4     4     4     2     9     5     1     1

Average Waiting Time    : 2.50
Average Turnaround Time : 6.25
Average Response Time   : 2.50
CPU Busy Time           : 15
CPU Idle Time           : 0
CPU Utilization         : 100.00%


========== PRIORITY PREEMPTIVE ==========

========== GANTT CHART ==========
| P1 (0-1) | P2 (1-7) | P4 (7-11) | P1 (11-13) | P3 (13-15) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     3     3     13    13    10    0
P2     1     6     1     7     6     0     0
P3     2     2     4     15    13    11    11
P4     4     4     2     11    7     3     3

Average Waiting Time    : 6.00
Average Turnaround Time : 9.75
Average Response Time   : 3.50
CPU Busy Time           : 15
CPU Idle Time           : 0
CPU Utilization         : 100.00%

Enter Time Quantum for Round Robin: 2


========== ROUND ROBIN ==========

========== GANTT CHART ==========
| P1 (0-2) | P2 (2-4) | P3 (4-6) | P1 (6-7) | P4 (7-9) | P2 (9-11) | P4 (11-13) | P2 (13
-15) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     3     3     7     7     4     0
P2     1     6     1     15    14    8     1
P3     2     2     4     6     4     2     2
P4     4     4     2     13    9     5     3

Average Waiting Time    : 4.75
Average Turnaround Time : 8.50
Average Response Time   : 1.50
CPU Busy Time           : 15
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

N

Figure 14: Overall Scheduling Algorithm Comparison for Test Case 2.

Test Case 3 — Process Scheduling Simulation

Input Details

Process ID	Arrival Time	Burst Time	Priority
P1	0	5	2
P2	1	3	3
P3	3	6	1
P4	5	2	4

Round Robin Time Quantum: 2

1. FCFS (First Come, First Served)

```
========== FCFS ==========

========== GANTT CHART ==========
| P1 (0-5) | P2 (5-8) | P4 (8-10) | P3 (10-11) |

PID    AT    BT    PR    CT    TAT    WT    RT
P1     0     5     2     5     5      0     0
P2     1     3     3     8     7      4     4
P3     6     1     2     11    5      4     4
P4     5     2     4     10    5      3     3

Average Waiting Time    : 2.75
Average Turnaround Time : 5.50
Average Response Time   : 2.75
CPU Busy Time           : 11
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

Figure 3.1 FCFS Gantt Chart — Test Case 3

2. SJF Non-Preemptive

```
========== SJF NON-PREEMPTIVE ==========

========== GANTT CHART ==========
| P1 (0-5) | P4 (5-7) | P3 (7-8) | P2 (8-11) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     5     2     5     5     0     0
P2     1     3     3     11    10    7     7
P3     6     1     2     8     2     1     1
P4     5     2     4     7     2     0     0

Average Waiting Time    : 2.00
Average Turnaround Time : 4.75
Average Response Time   : 2.00
CPU Busy Time           : 11
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

Figure 3.2 SJF Non-Preemptive Gantt Chart — Test Case 3

3. Priority Non-Preemptive Scheduling

```
========== PRIORITY NON-PREEMPTIVE ==========

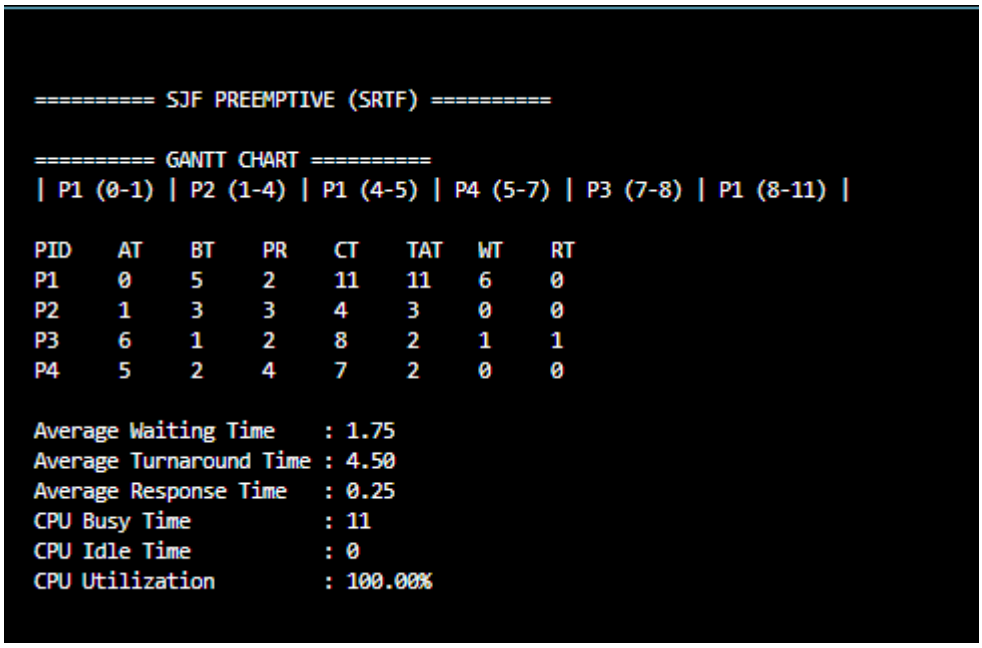
========== GANTT CHART ==========
| P1 (0-5) | P2 (5-8) | P3 (8-9) | P4 (9-11) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     5     2     5     5     0     0
P2     1     3     3     8     7     4     4
P3     6     1     2     9     3     2     2
P4     5     2     4     11    6     4     4

Average Waiting Time    : 2.50
Average Turnaround Time : 5.25
Average Response Time   : 2.50
CPU Busy Time           : 11
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

Figure 3.3 Priority Non-Preemptive Gantt Chart — Test Case 3

4. SJF Preemptive (Shortest Remaining Time First — SRTF)



```
========== SJF PREEMPTIVE (SRTF) ==========

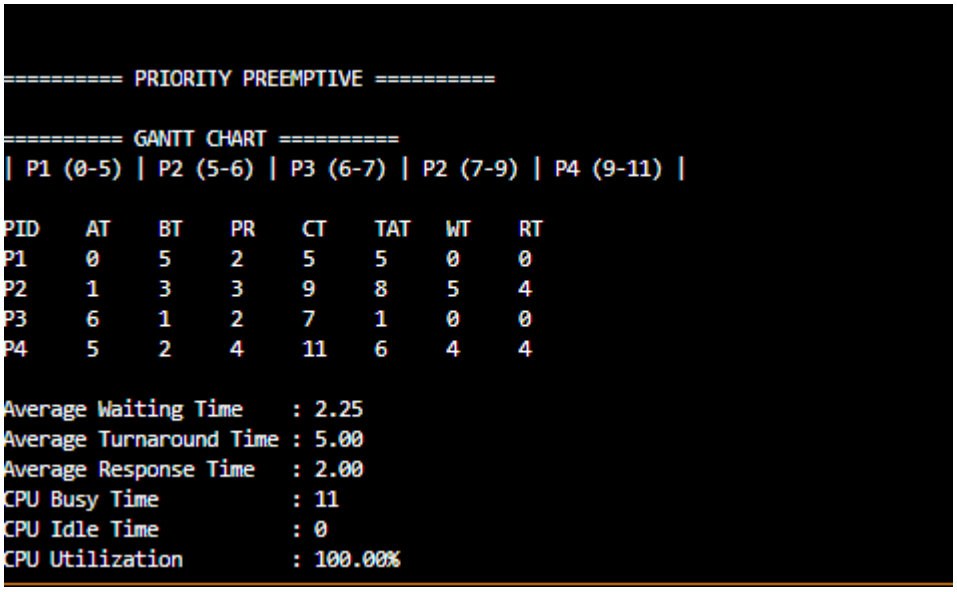
========== GANTT CHART ==========
| P1 (0-1) | P2 (1-4) | P1 (4-5) | P4 (5-7) | P3 (7-8) | P1 (8-11) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     5     2     11    11    6     0
P2     1     3     3     4     3     0     0
P3     6     1     2     8     2     1     1
P4     5     2     4     7     2     0     0

Average Waiting Time    : 1.75
Average Turnaround Time : 4.50
Average Response Time   : 0.25
CPU Busy Time           : 11
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

Figure 3.4 SJF Preemptive (SRTF) Gantt Chart — Test Case 3

5. Priority Preemptive Scheduling



```
========== PRIORITY PREEMPTIVE ==========

========== GANTT CHART ==========
| P1 (0-5) | P2 (5-6) | P3 (6-7) | P2 (7-9) | P4 (9-11) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     5     2     5     5     0     0
P2     1     3     3     9     8     5     4
P3     6     1     2     7     1     0     0
P4     5     2     4     11    6     4     4

Average Waiting Time    : 2.25
Average Turnaround Time : 5.00
Average Response Time   : 2.00
CPU Busy Time           : 11
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

Figure 3.5 Priority Preemptive Gantt Chart — Test Case 3

6. Round Robin Scheduling

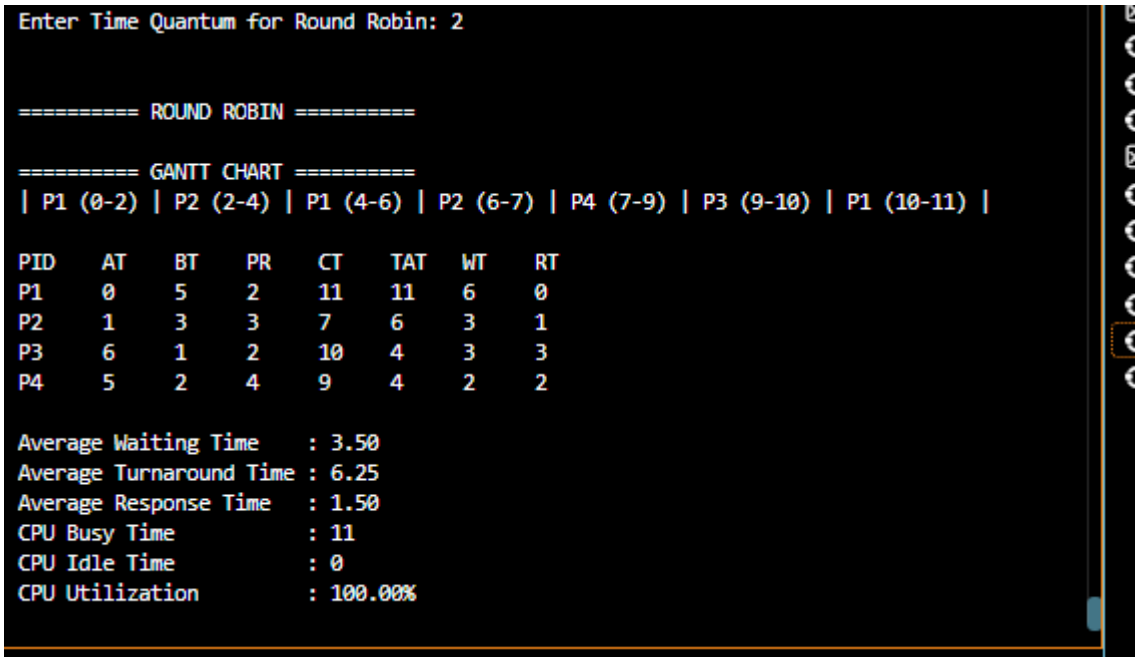


Figure 3.6 Round Robin Gantt Chart — Test Case 3

7. Test Case 3 — Algorithm ComparisoN

```
========== PRIORITY NON-PREEMPTIVE ==========

========== GANTT CHART ==========
| P1 (0-5) | P2 (5-8) | P3 (8-9) | P4 (9-11) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     5     2     5     5     0     0
P2     1     3     3     8     7     4     4
P3     6     1     2     9     3     2     2
P4     5     2     4     11    6     4     4

Average Waiting Time    : 2.50
Average Turnaround Time : 5.25
Average Response Time   : 2.50
CPU Busy Time           : 11
CPU Idle Time           : 0
CPU Utilization         : 100.00%


========== SJF PREEMPTIVE (SRTF) ==========

========== GANTT CHART ==========
| P1 (0-1) | P2 (1-4) | P1 (4-5) | P4 (5-7) | P3 (7-8) | P1 (8-11) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     5     2     11    11    6     0
P2     1     3     3     4     3     0     0
P3     6     1     2     8     2     1     1
P4     5     2     4     7     2     0     0

Average Waiting Time    : 1.75
Average Turnaround Time : 4.50
Average Response Time   : 0.25
CPU Busy Time           : 11
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

```
========== PRIORITY PREEMPTIVE ==========

========== GANTT CHART ==========
| P1 (0-5) | P2 (5-6) | P3 (6-7) | P2 (7-9) | P4 (9-11) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     5     2     5     5     0     0
P2     1     3     3     9     8     5     4
P3     6     1     2     7     1     0     0
P4     5     2     4     11    6     4     4

Average Waiting Time    : 2.25
Average Turnaround Time : 5.00
Average Response Time   : 2.00
CPU Busy Time           : 11
CPU Idle Time           : 0
CPU Utilization         : 100.00%

Enter Time Quantum for Round Robin: 2


========== ROUND ROBIN ==========

========== GANTT CHART ==========
| P1 (0-2) | P2 (2-4) | P1 (4-6) | P2 (6-7) | P4 (7-9) | P3 (9-10) | P1 (10-11) |

PID    AT    BT    PR    CT    TAT   WT    RT
P1     0     5     2     11    11    6     0
P2     1     3     3     7     6     3     1
P3     6     1     2     10    4     3     3
P4     5     2     4     9     4     2     2

Average Waiting Time    : 3.50
Average Turnaround Time : 6.25
Average Response Time   : 1.50
CPU Busy Time           : 11
CPU Idle Time           : 0
CPU Utilization         : 100.00%
```

1. Overall Performance Comparison

The six CPU scheduling algorithms were compared using Average Waiting Time, Average Turnaround Time, Average Response Time, and CPU Utilization. The comparison was performed using Test Case 3 with four processes and a Round Robin time quantum of 2.

Scheduling Algorithm	Avg. Waiting Time	Avg. Turnaround Time	Avg. Response Time	CPU Utilization
FCFS	2.75	5.50	2.75	100%
SJF Non-Preemptive	2.00	4.75	2.00	100%
Priority Non-Preemptive	2.50	5.25	2.50	100%
SJF Preemptive (SRTF)	1.75	4.50	0.25	100%
Priority Preemptive	2.25	5.00	2.00	100%
Round Robin (Q = 2)	3.50	6.25	1.50	100%

The results show that SJF Preemptive (SRTF) achieved the lowest average waiting time, turnaround time, and response time for this test case. Round Robin had the highest average waiting and turnaround times among the six algorithms. All algorithms achieved 100% CPU utilization for this workload because the CPU had no idle intervals.

Performance Analysis

The performance analysis shows that different CPU scheduling algorithms produce different results depending on process arrival times, burst times, and priorities.

FCFS schedules processes in the order they arrive and is simple to implement, but shorter processes may wait behind longer processes. SJF Non-Preemptive reduces waiting time by selecting the shortest available process, while Priority Non-Preemptive selects processes according to priority.

SJF Preemptive, also known as Shortest Remaining Time First (SRTF), achieved the best overall average metrics in Test Case 3 because it can interrupt a running process when a process with a shorter remaining burst time becomes available. Priority Preemptive can also interrupt a running process when a higher-priority process arrives.

Round Robin gives each ready process a fixed time quantum, allowing processes to share CPU time. Its performance depends on the selected quantum. In this test case, a quantum of 2 produced a higher average waiting time and turnaround time than the other algorithms.

CPU utilization was 100% for all six algorithms in this test case. This result applies to the tested workload and does not mean that every algorithm will achieve 100% utilization for every possible input.

Conclusion

The Process Scheduling Simulator was implemented in C to demonstrate and compare six CPU scheduling algorithms: FCFS, SJF Non-Preemptive, Priority Non-Preemptive, SJF Preemptive (SRTF), Priority Preemptive, and Round Robin.

The simulator accepts process details such as PID, arrival time, burst time, and priority, along with a time quantum for Round Robin. It displays Gantt charts and calculates completion time, turnaround time, waiting time, response time, average performance metrics, and CPU utilization.

The test results demonstrate that no single scheduling algorithm is best for every situation. In Test Case 3, SJF Preemptive achieved the lowest average waiting, turnaround, and response times. The project helped us understand how scheduling policies affect process execution and system performance.

Overall, this project provides a practical understanding of CPU scheduling and demonstrates how scheduling algorithms can be evaluated using performance metrics.